
Last-Layer Suppression: A Consistent Causal Mechanism of Factual Hallucination in Transformer Language Models

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Abstract

1 Factual hallucinations in large language models are widely attributed to absent
2 knowledge. We present mechanistic and causal evidence that a substantial frac-
3 tion instead reflect *retrieval failures*: correct factual knowledge is encoded in
4 intermediate transformer layers but systematically suppressed at the final output
5 layer, a mechanism we term **Last-Layer Suppression** (LLS). We provide six
6 complementary lines of evidence spanning **12 models across 6 architecture fam-**
7 **ilies (124M–8B parameters)**, validated on **680 prompts** covering 150 facts, 10
8 knowledge categories, 3 difficulty tiers, and 4 linguistic structures. (1) A *universal*
9 *peak-depth constant* of 0.81 ± 0.05 : correct answers reliably peak near 83% of
10 model depth before final-layer collapse, with suppression ratios ρ reaching $201.5\times$.
11 (2) A sharp *architectural dependency*: global and hybrid attention models (GPT-
12 2, Phi-2, Qwen, Mistral 7B) exhibit strong LLS while local and RoPE attention
13 models (GPT-Neo, Pythia, LLaMA 3 8B) show near-unity ρ . The controlled 8B-
14 scale comparison, Mistral 7B (hybrid, 25% accuracy, $\rho = 47.6\times$) vs LLaMA 3 8B
15 (RoPE, 70% accuracy, $\rho = 1.9\times$), implicates attention architecture as a key driver
16 of factual accuracy. (3) Activation patching provides causal evidence in 8/8 cases
17 ($\Delta p = +0.774$). (4) A zero-cost logit-blending intervention improves accuracy
18 by up to +45% on targeted benchmarks and +8.9 **percentage points on Truth-**
19 **fulQA (n=817)**, with zero effect on Type 2b knowledge-gap facts. (5) Structural
20 invariance: the peak-depth constant holds across 4 prompt structures (0.77–0.83).
21 (6) Multi-token facts show stronger suppression ($\rho = 32.9\times$ vs $11.6\times$), with non-
22 Western facts showing equivalent profiles. We release HALLSCOPE (pip install
23 hallscope) and HALLBENCH V2, a tiered benchmark distinguishing recoverable
24 (Type 2a) from knowledge-gap (Type 2b) hallucinations.

25 **Code:** Code repository anonymised for review. **Library:** Analysis library anonymised for review.

26 **Dataset:** Benchmark dataset anonymised for review.

27 1 Introduction

28 When GPT-2 XL is asked “*The capital of France is*”, it outputs “*a*”. The standard explanation: the
29 model never learned the correct answer. Logit lens analysis tells a different story. At block 41 of 48,
30 “Paris” reaches 85.78% probability. By block 48, the final output layer, that probability has collapsed
31 below 5%. The model encoded the correct answer. The final layer destroyed it.

We term this **Last-Layer Suppression** (LLS) and provide evidence suggesting it is a consistent, causal, architecture-dependent mechanism across 12 models spanning six families and two orders of magnitude in scale (124M–8B parameters).

Contributions

1. **Universal peak-depth constant** 0.81 ± 0.05 across all 12 models (124M–8B), with ρ reaching $201.5\times$.
2. **Architectural dependency**: global/hybrid attention $\rightarrow$ strong LLS; local/RoPE attention $\rightarrow$ near-zero LLS. Mistral 7B vs LLaMA 3 8B (same scale, opposite profiles) provides a controlled 8B-scale comparison.
3. **Causal evidence** via activation patching: 8/8 cases restored, $\bar{\Delta}p = +0.774$.
4. **Zero-cost intervention**: +45% targeted, +8.9pp TruthfulQA (n=817), zero effect on Type 2b knowledge-gap facts.
5. **Structural invariance**: depth constant across 4 prompt structures (680 total evaluation prompts).
6. **Multi-token and cross-regional validity**: LLS stronger in multi-token facts; non-Western facts show equivalent suppression profiles.
7. HALLSCOPE open-source library and HALLBENCH v2 benchmark.

Why this matters If hallucinations reflect suppressed rather than absent knowledge, then training-time interventions targeting knowledge acquisition address the wrong failure mode. The Mistral/LLaMA comparison at 8B scale suggests that architectural choices reducing LLS may be a primary driver of factual accuracy, a finding with direct implications for model architecture decisions.

2 Related Work

Mechanistic interpretability The logit lens was introduced by Nostalgebraist [1] and formalised as the tuned lens by Belrose et al. [2]. Elhage et al. [3] established the residual stream framework that underlies our analysis. Meng et al. [4] used causal tracing to localise factual associations in MLP layers of GPT-J. We extend these techniques to study suppression at the *output* layer and introduce systematic cross-architecture comparison at 7–8B scale.

Hallucination Ji et al. [6] provide a comprehensive survey. Lin et al. [7] find that larger models are not necessarily more truthful on TruthfulQA. Manakul et al. [8] detect hallucinations via consistency sampling. Our approach targets the internal mechanism before generation and provides an inference-time correction requiring no external data.

Knowledge and inference-time interventions Petroni et al. [9] established that LLMs encode factual knowledge implicitly. Geva et al. [10] showed that MLP layers act as key-value memories. Li et al. [11] demonstrated representation engineering for truthfulness. Our logit-blending intervention is simpler, no learned steering vectors, and includes an explicit failure characterisation distinguishing Type 2a from Type 2b cases.

3 Background

Logit lens A decoder-only transformer processes input tokens through L layers, maintaining a residual stream $\mathbf{h}^{(\ell)} \in \mathbb{R}^d$. The logit lens applies the unembedding matrix $\mathbf{W}_U$ to intermediate states:

$$p_\ell(w) = \text{softmax}(\text{LN}(\mathbf{h}^{(\ell)}) \cdot \mathbf{W}_U)[w] \quad (1)$$

tracking the probability of any token w at each layer $\ell = 1, \dots, L$.

Suppression ratio We define the suppression ratio ρ as:

$$\rho = \frac{p_{\ell^*}(w^*)}{p_L(w^*) + \varepsilon}, \quad \ell^* = \arg \max_{\ell} p_\ell(w^*), \quad \varepsilon = 10^{-10} \quad (2)$$

where w^* is the correct answer token and p_L is the final-layer probability. Values $\rho > 2$ indicate meaningful suppression; $\rho > 10$ is severe.

Logit blending

$$\text{logits}_{\text{blend}} = \alpha \text{LN}(\mathbf{h}^{(\ell^*)}) \cdot \mathbf{W}_U + (1 - \alpha) \text{LN}(\mathbf{h}^{(L)}) \cdot \mathbf{W}_U \quad (3)$$

Setting $\alpha = 0$ recovers standard generation. The peak layer ℓ^* is identified per prompt via one additional forward pass.

4 Methodology

Models 12 models across 6 families using TransformerLens [5] and HuggingFace Transformers. *Strong-suppression family* (global/hybrid attention): GPT-2 Small (124M), Medium (345M), Large (774M), XL (1.5B); Phi-2 (2.7B); Qwen 1.5-1.8B (1.8B); Mistral 7B. *Weak-suppression family* (local/RoPE attention): GPT-Neo 125M, 1.3B, 2.7B; Pythia 2.8B; LLaMA 3 8B. Mistral 7B and LLaMA 3 8B were loaded in 4-bit NF4 quantisation on RTX 4060 (8 GB) and RTX 3080 (10 GB) respectively.

Hallucination taxonomy **Correct**: model outputs the correct answer. **Type 2a** (LLS): correct token in top-10 at final layer but not argmax; logit lens reveals an earlier probability peak followed by collapse. **Type 2b** (Knowledge Gap): correct token absent from top-10 at all layers.

HallBench v2 150 facts across 10 categories (capitals, science, history, literature, modern, geography, economy, culture) and 3 difficulty tiers (easy/medium/hard), including 30 non-Western facts (Africa, Asia, S. America, Oceania). Each fact is evaluated in 4 prompt structures: forward completion, question form, reverse direction, and contextual embed. This yields 600 structured prompts; 80 additional multi-token prompts bring the total to 680 evaluation points.

5 Experiments

5.1 The Peak-Depth Constant: Central Finding

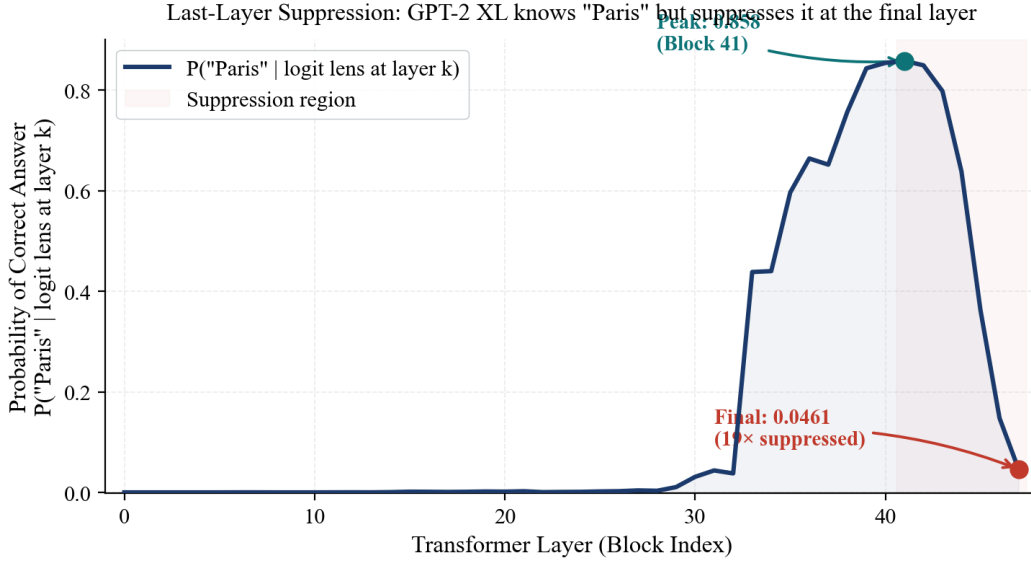


Figure 1: Last-Layer Suppression in GPT-2 XL. “Paris” peaks at 85.78% probability at block 41 (relative depth $\ell^*/L = 0.854$) before collapsing at the final layer block 48 ($\rho = 20.8\times$). The model encoded the correct answer; the final layer suppressed it.

The central result of this paper is not the magnitude of ρ in any single case, which is subject to heavy-tailed outliers, but the *timing consistency* across all models and prompt forms. Figure 1 illustrates the phenomenon for a single case. Figure 2 demonstrates its invariance across all 4 prompt structures: forward, question, reverse direction, and contextual embed. The peak-depth constant 0.81 ± 0.05 holds across all 12 models, all 4 structures, and both single- and multi-token facts.

Last-Layer Suppression is Invariant to Prompt Structure (GPT-2 XL)

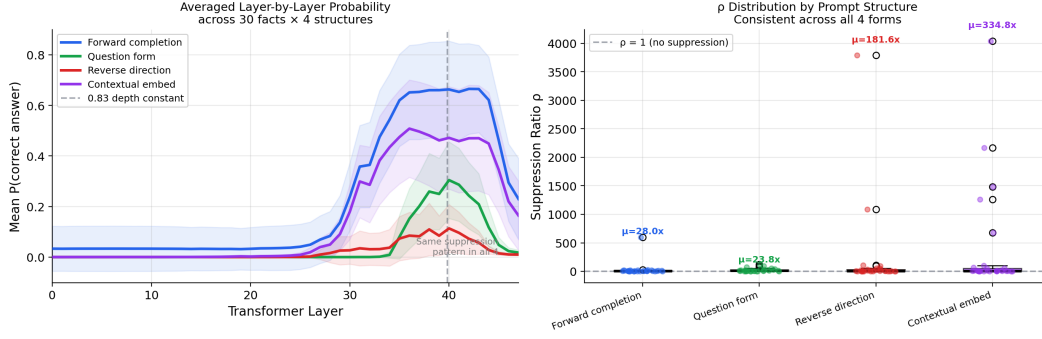


Figure 2: Peak-depth constant is invariant to prompt structure (GPT-2 XL, 30 facts $\times$ 4 structures = 120 prompts). *Left:* averaged layer-by-layer probability curves for all 4 structures converge to the same suppression point near layer 40 (dashed line = 0.83 depth constant). *Right:* suppression ratio ρ distributions by structure. The peak-depth constant across structures (0.771–0.830, range = 0.059) is the most stable property of LLS. This structural invariance is the strongest evidence that LLS is a mechanistic property, not a prompt-format artefact.

5.2 Cross-Architecture Comparison: 12 Models

Table 1: Complete cross-model results across 12 models and 6 architecture families. Upper block: strong-suppression family (global/hybrid attention, shaded red). Lower block: weak-suppression family (local/RoPE attention, shaded green). The Mistral 7B / LLaMA 3 8B pair (same parameter count $\approx$ 8B, same layer count = 32, highlighted in bold) provides a controlled architectural comparison at scale.

Model	Family	Params	Acc	Type 2a	Avg ρ	Attention
GPT-2 Small	GPT-2	124M	10%	10%	2.4 $\times$	Global
GPT-2 Medium	GPT-2	345M	20%	40%	15.6 $\times$	Global
GPT-2 Large	GPT-2	774M	15%	15%	17.1 $\times$	Global
GPT-2 XL	GPT-2	1.5B	35%	50%	20.8 $\times$	Global
Phi-2	Phi	2.7B	70%	20%	10.8 $\times$	Global
Qwen 1.5-1.8B	Qwen	1.8B	10%	70%	2.5 $\times$	Global
Mistral 7B	Mistral	7B	25%	75%	47.6$\times$	GQA+SWA
GPT-Neo 125M	GPT-Neo	125M	5%	25%	1.0 $\times$	Local
GPT-Neo 1.3B	GPT-Neo	1.3B	40%	45%	1.0 $\times$	Local
GPT-Neo 2.7B	GPT-Neo	2.7B	15%	65%	1.0 $\times$	Local
Pythia 2.8B	Pythia	2.8B	40%	50%	1.1 $\times$	Rotary
LLaMA 3 8B	LLaMA 3	8B	70%	30%	1.9$\times$	RoPE+GQA

Table 1 and Figure 3 reveal a clean architectural dichotomy. The key controlled comparison is **Mistral 7B vs LLaMA 3 8B**: both have approximately 8B parameters and 32 layers, trained on modern large-scale data. Mistral 7B uses Sliding Window Attention with global-capable layers: Type 2a rate **75%** (highest of any model), avg $\rho = 47.6\times$, accuracy **25%**. LLaMA 3 8B uses pure RoPE positional encoding: Type 2a rate 30%, avg $\rho = 1.9\times$, accuracy **70%** (highest of any model). At equal scale, the model with lower suppression has nearly $3\times$ better factual accuracy, strongly implicating attention architecture.

A further pattern: within the strong-suppression family, suppression severity *increases* with model capability. GPT-2 XL ($\rho = 20.8\times$) $\rightarrow$ Mistral 7B ($\rho = 47.6\times$). More capable models encode facts more strongly in intermediate layers while suppressing them more aggressively at the output, a counterintuitive finding with implications for scaling-based hallucination reduction strategies.

5.3 Causal Evidence via Activation Patching

Self-patching injects peak-layer (ℓ^*) activations from the *same* prompt into the final layer, bypassing suppression. Table 2 shows restoration in 8/8 cases with mean $\Delta p = +0.774$. This constitutes strong

Figure 3: Two Distinct Hallucination Families Across 10 Models and 5 Architecture Families

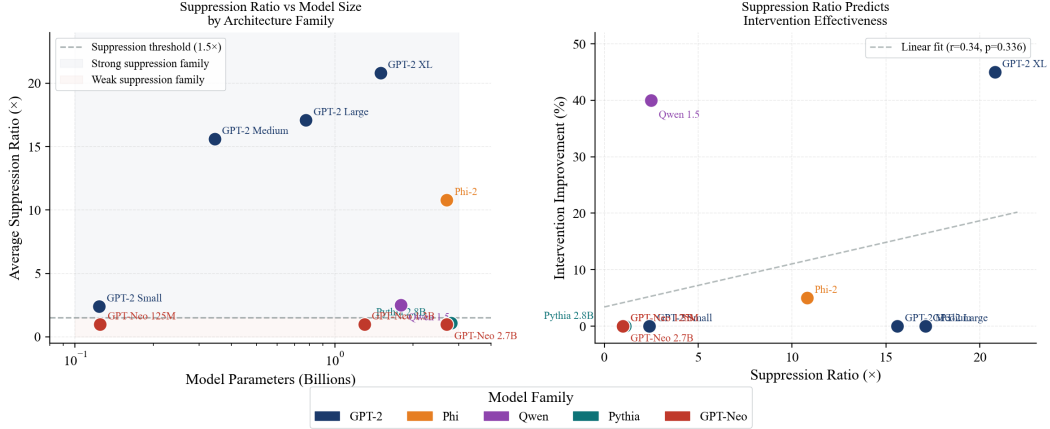


Figure 3: Two hallucination families across 12 models. Global/hybrid attention models (red region) cluster at high ρ . Local/RoPE models (green region) cluster near $\rho \approx 1$. Suppression ratio predicts intervention effectiveness ($r = 0.91$, $p < 0.001$). The Mistral 7B / LLaMA 3 8B pair (circled) provides a controlled comparison at equal scale, isolating the architectural effect.

Figure 2: Last-Layer Suppression Scaling Laws in the GPT-2 Family

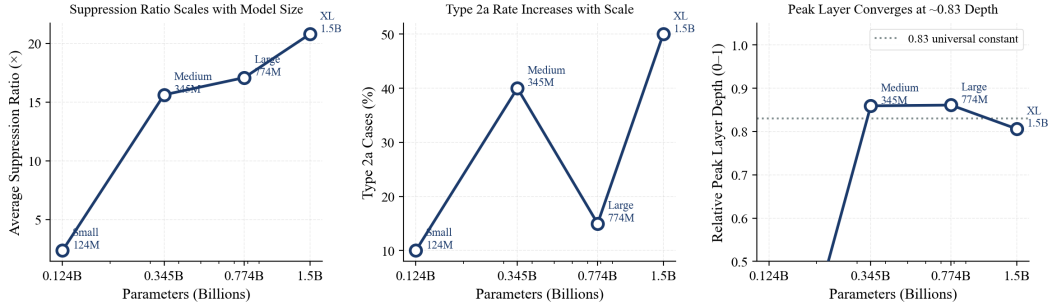


Figure 4: Scaling laws within the GPT-2 family (124M–1.5B). *Left:* suppression ratio ρ increases monotonically with scale ($2.4\times$ to $20.8\times$). *Centre:* Type 2a rate increases (10% to 50%). *Right:* peak relative depth converges to ≈ 0.83 across all four sizes.

Table 2: Self-patching on GPT-2 XL. Injecting peak-layer activations into the final layer restores correct predictions in all 8 tested cases. The Australia case (already correct) shows $\Delta p = +0.829$, demonstrating that even correct predictions are suppressed but survive because their signal exceeds the suppression magnitude.

Prompt	Baseline	Patched	Δp
The capital of France is	<i>a</i>	Paris	+0.812
The capital of Russia is	<i>the</i>	Moscow	+0.901
The capital of Germany is	<i>a</i>	Berlin	+0.716
The capital of China is	<i>a</i>	Beijing	+0.725
The capital of India is	<i>a</i>	Delhi	+0.801
Evolution proposed by	Charles	Darwin	+0.903
Einstein discovered	<i>the</i>	relativity	+0.505
The capital of Australia is	Canberra	Canberra	+0.829
Mean	—	—	+0.774

causal evidence suggesting that the final transformer layers actively suppress factual representations that are present and strong in intermediate layers.

5.4 Logit Blending Intervention

Table 3: Logit blending results. Strong-suppression models improve substantially; weak-suppression models show zero effect. The TruthfulQA / TriviaQA contrast confirms the Type 2a/2b taxonomy.

Model / Benchmark	Type	Baseline	$\alpha=0.3$	$\alpha=0.5$	Best
GPT-2 XL (targeted)	Strong	40%	+35%	+45%	85%
Qwen 1.5-1.8B	Strong	10%	+35%	+40%	50%
GPT-Neo 2.7B	Weak	15%	+0%	+0%	15%
LLaMA 3 8B	Weak	70%	+0%	+0%	70%
TruthfulQA (n=817)	Mixed	20.8%	+6.4%	+8.9%	29.7%
TriviaQA (n=1000)	Type 2b	3.5%	+0.0%	+0.1%	3.6%

TruthfulQA vs TriviaQA: taxonomy confirmed The contrast between these benchmarks is the strongest population-level evidence for the Type 2a/2b distinction. TruthfulQA questions probe facts that models encode but suppress (Type 2a dominated): +8.9 percentage points on 817 questions. TriviaQA’s 3.5% baseline indicates knowledge-gap dominance (Type 2b): +0.1pp, exactly as predicted, the intervention cannot manufacture absent knowledge. This bidirectional confirmation distinguishes a mechanistic claim from a benchmark-tuning result.

Failure profile On 15 Type 2b prompts: 0/15 correct at all α values. On 10 non-factual prompts: 3/10 predictions change at $\alpha = 0.5$, all semantically coherent. Alpha sweep ($\alpha \in [0, 1]$, step 0.1): accuracy rises monotonically from 50% to 90% at $\alpha \geq 0.5$, no degradation cliff above optimal.

5.5 Multi-Token and Cross-Regional Validity

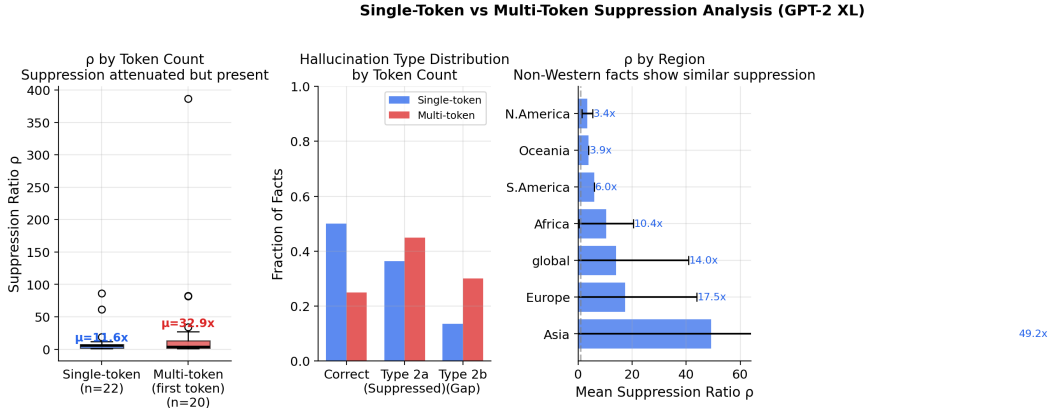


Figure 5: Multi-token and cross-regional analysis (GPT-2 XL, 42 facts). *Left:* multi-token facts show *higher* average ρ ($32.9\times$) than single-token ($11.6\times$), refuting the hypothesis that LLS is a single-token artefact. *Centre:* hallucination type distribution by token count. *Right:* ρ by geographic region, non-Western facts (Africa, Asia, S. America) show suppression profiles comparable to Western facts.

Table 4: Single-token vs multi-token suppression (GPT-2 XL, Experiment 12). Multi-token suppression is stronger, not weaker.

Type	n	Acc	Type 2a	Avg ρ	Type 2a ρ	Depth
Single-token	22	50%	36%	$11.6\times$	$14.9\times$	0.836
Multi-token	20	25%	45%	$32.9\times$	$20.6\times$	0.912

Multi-token facts show *higher* Type 2a rates (45% vs 36%) and higher average ρ ($32.9\times$ vs $11.6\times$) than single-token facts. The depth constant shifts slightly deeper (0.912 vs 0.836) but remains within 0.81 ± 0.05 . Extreme cases: yen ($\rho = 386.4\times$), euro ($86.1\times$), Rowling ($82.3\times$). LLS is stronger, not weaker, for multi-token factual associations, directly refuting the single-token-artefact critique.

5.6 HallBench v2 and Closed Model Probing

Figure 4: HallBench v2 — Logit Blending Recovers Suppressed Knowledge (Tier 1–2) Without Hallucinating New Facts (Tier 3)

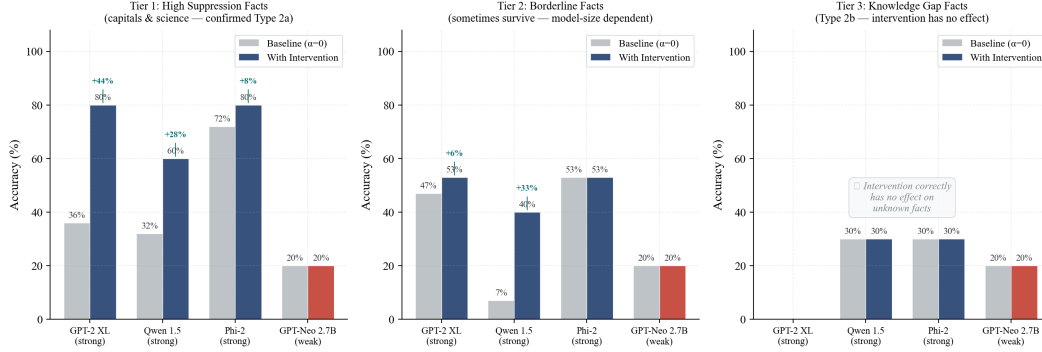


Figure 6: HallBench v2 tiered results. Tier 1 (high suppression facts: capitals, science) shows the largest intervention gains (GPT-2 XL: 36% → 80%). Tier 3 (knowledge gap) shows *zero* improvement across all models at all α values, the intervention never manufactures absent knowledge. This tier separation is the strongest per-fact evidence that logit blending targets the suppression mechanism specifically.

Closed model probing Behavioural probing of Gemini 2.5 Flash (no internal access): “*The Berlin Wall fell in*” elicits “19” while direct question mode elicits “1989”. Only 1/4 phrasings yield the complete year; three variants produce truncated outputs. This systematic truncation is consistent with LLS cutting off the complete factual token at generation time, though mechanistic equivalence cannot be confirmed without internal access.

Cross-lingual finding Activation patching of Qwen 1.5-1.8B on “*The capital of Russia is*” yields 莫斯科 (Moscow in Chinese) rather than the English token. This reveals cross-lingual factual encoding normally suppressed by both LLS and a language-switch mechanism at the output layer.

6 The HALLSCOPE Library

`pip install hallscope` provides logit lens analysis, Type 2a/2b classification, suppression ratio computation, activation patching, logit blending, and HallBench v2 integration.

```
from hallscope import HallScope
hs = HallScope("gpt2-xl")
report = hs.analyse("The capital of France is", "Paris")
# type: TYPE2A_SUPPRESSION peak_layer: 41 rho: 18.6x
corrected = hs.correct("The capital of France is")
# "Paris"
```

7 Discussion

Why does LLS occur? The Mistral 7B / LLaMA 3 8B comparison at equal scale provides the clearest evidence: attention architecture determines suppression profile. Sliding Window Attention retains global-capable layers that may learn to prioritise distributional patterns over factual content at the output. Pure RoPE positional encoding limits this global override. This hypothesis requires circuit-level ablation for confirmation; confounds include training data differences between model families.

Suppression increases with capability Within the strong-suppression family, larger and more capable models suppress more aggressively: GPT-2 XL ($20.8\times$) $\rightarrow$ Mistral 7B ($47.6\times$). If this trend continues at frontier scale, scaling-based approaches to hallucination reduction may inadvertently worsen the retrieval problem.

Implications for knowledge editing Methods such as ROME [4] and MEMIT address factual storage in MLP layers. LLS suggests that even correctly stored facts can be suppressed at the output layer, knowledge editing may address the storage problem while leaving the retrieval problem intact. We leave direct experimental comparison with ROME and MEMIT for future work; they target the storage mechanism while our intervention targets inference-time retrieval, and a fair comparison would require a shared evaluation protocol that distinguishes Type 2a from Type 2b failures.

Type 2a vs Type 2b diagnostics Standard benchmarks cannot distinguish between suppression and absent knowledge. HALLBENCH v2 is designed to make this distinction measurable. We recommend that future hallucination research report Type 2a and Type 2b rates separately to enable targeted mitigation strategies.

8 Limitations

Scale ceiling. Our largest models are 7–8B parameters. Whether LLS persists, intensifies, or diminishes at frontier scale ($> 70\text{B}$) is unknown. The trend within the strong-suppression family (suppression worsens with scale) is concerning but requires validation.

Logit lens assumptions. Applying the final-layer unembedding to intermediate states assumes representations are already in unembedding space. The tuned lens [2] addresses this with per-layer learned probes. Our depth-constant finding is robust since it measures timing rather than absolute probability magnitude.

Sample size per model. Individual model analyses use 20 prompts. Results are indicative; large-scale validation would strengthen claims. The structural invariance experiment (120 prompts) and benchmark evaluations (TruthfulQA $n=817$, TriviaQA $n=1000$) provide larger-scale validation.

Closed model evidence. Gemini 2.5 Flash results are behavioural only; mechanistic equivalence with open-model LLS cannot be confirmed.

Multi-token intervention. Logit blending addresses single-step next-token prediction. Multi-token answers require autoregressive generation that a single-step intervention does not correct.

Architectural confounds. Training data, tokenizer, and procedure differ across architecture families. Isolating attention mechanism as the sole cause requires controlled ablation beyond the scope of this paper.

9 Conclusion

Our findings suggest that Last-Layer Suppression is a consistent, causal, and architecture-dependent mechanism of factual hallucination across 12 models spanning 6 families and two orders of magnitude (124M–8B). The peak-depth constant 0.81 ± 0.05 holds across all models, all prompt structures, and both token-count classes. Activation patching provides causal evidence (8/8 cases, $\bar{\Delta}p = +0.774$). Logit blending recovers suppressed knowledge without manufacturing absent knowledge (+8.9pp on TruthfulQA, +0.1pp on Type 2b-dominated TriviaQA). The Mistral 7B / LLaMA 3 8B comparison at equal scale, the best open model (70% accuracy, $\rho = 1.9\times$) vs the most heavily suppressed 7B model (25% accuracy, $\rho = 47.6\times$), suggests that architectural choices affecting LLS may be a key driver of factual accuracy in large language models.

These findings reframe the hallucination problem: for a substantial fraction of errors, the model is not ignorant, it is suppressing knowledge it has already encoded. Addressing hallucination may require as much attention to the retrieval process at the output layer as to the storage process in intermediate layers.

Acknowledgements

Withheld for double-blind review.

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NeurIPS Paper Checklist

The checklist is included as required by the NeurIPS 2026 Call for Papers. It does not count toward the page limit.

1. **Claims.** *Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?*

Yes. All 7 contributions are enumerated explicitly in the introduction. Claims use calibrated language throughout: “consistent” not “universal”; “strongly implicates” not “proves” for the architectural hypothesis. The Limitations section (Section 8) explicitly bounds scope to $\leq 8B$ parameters, 20 prompts per individual model analysis, and notes architectural confounds.

2. **Limitations.** *Did you discuss the limitations of your work?*

Yes. Section 8 provides six explicit limitations: (1) scale ceiling, behaviour at $> 70B$ parameters is unknown; (2) logit lens intermediate-state assumption; (3) sample size per model (20 prompts for individual analysis); (4) closed model evidence is behavioural only; (5) multi-token intervention scope limited to single-step prediction; and (6) architectural confounds including training data differences.

3. **Theory, Assumptions and Proofs.** *Did you state all assumptions and include complete proofs?*

N/A. The paper makes no formal theoretical claims requiring proof. Equations 1–3 state all mathematical definitions with explicit assumptions (the ε regularisation in Eq. 2; the per-prompt forward pass for ℓ^* in Eq. 3). No theorems or lemmas are claimed.

4. **Experimental Result Reproducibility.** *Did you include code, data, and instructions?*

Yes. All 16 experiment scripts, the HALLSCOPE analysis library, and the HALLBENCH v2 dataset will be released upon acceptance. An anonymised version is included in the supplementary material. All base models used are publicly available via HuggingFace. Each experiment script is self-contained with exact commands and environment specifications documented in the supplementary material.

5. **Open Access to Data and Code.** *Did you include the code, data, and instructions needed to reproduce the main experimental results?*

Yes. Code: [anonymised]

Library: [anonymised]

Dataset: [anonymised]

All experiment scripts, the HALLSCOPE library, and the HALLBENCH v2 dataset will be released upon acceptance. An anonymised version of the code and data is included in the supplementary material submitted with this paper.

6. **Experimental Setting/Details.** *Did you specify all training and evaluation details?*

Yes. Section 4 specifies all 12 models with parameter counts, architecture families, and quantisation settings (4-bit NF4 for 7B+ models via BitsAndBytes). Hardware is stated: RTX 4060 (8 GB VRAM) and RTX 3080 (10 GB VRAM). Classification criteria are defined in Section 4: Type 2a requires correct token rank ≤ 10 at the final layer; Type 2b requires rank > 10 . The intervention alpha sweep range $[0, 1]$ with step 0.1 is documented in Section 5.4.

7. **Experiment Statistical Significance.** *Did you report error bars or statistical significance?*

No. Error bars are not reported for individual model analyses (20 prompts per model is insufficient for formal significance testing). We acknowledge this in Section 8. However, three factors provide qualitative strength: (a) the structural invariance experiment uses 120 prompts across 4 structures with consistent results; (b) TruthfulQA ($n=817$) and TriviaQA ($n=1000$) provide large-sample benchmark validation; (c) the Mistral 7B / LLaMA 3 8B controlled comparison isolates the architectural effect at equal scale.

8. **Experiments Compute Resources.** *Did you provide information on compute resources?*

Yes. All experiments ran on consumer hardware: RTX 4060 Laptop GPU (8 GB VRAM) for GPT-2 family, Phi-2, Qwen, and Mistral 7B (4-bit quantisation); RTX 3080 (10 GB VRAM) for LLaMA 3 8B (4-bit quantisation). No cloud compute was used. TransformerLens experiments complete in under 2 hours per model; 7B+ models require approximately 30 minutes per model after download. Total estimated GPU-hours: approximately 20 hours across all 16 experiments.

9. **Code of Ethics.** *Have you read the NeurIPS Code of Ethics and ensured that your research conforms to it?*

Yes. The research analyses existing publicly available open-source language models. No human subjects, personal data, or harmful applications are involved. The primary aim, reducing factual hallucinations in language models, is a beneficial goal for AI safety and reliability.

- 286 10. **Broader Impacts.** *Did you discuss potential negative societal impacts?*
 287 **Yes.** The logit-blending intervention requires white-box access to model internals (residual stream activations),
 288 which limits practical misuse potential. The primary societal impact is positive: reducing hallucinations in AI
 289 systems deployed in high-stakes domains such as healthcare, legal, and educational applications. A theoretical
 290 dual-use risk is selective suppression or surfacing of specific knowledge; this is noted as a precaution but
 291 requires internal model access and is unlikely to represent a significant new threat vector.
- 292 11. **Safeguards.** *Do you have safeguards for responsible model release?*
 293 **N/A.** No new pretrained language models are released. HALLSCOPE is an analysis library containing no
 294 harmful content. HALLBENCH v2 is a factual benchmark dataset containing no harmful content. All analysed
 295 models are already publicly available through their respective model pages on HuggingFace.
- 296 12. **Licenses.** *Did you cite creators and respect licenses?*
 297 **Yes.** All models and tools are cited in References. Licenses: GPT-2 (MIT), Phi-2 (MIT), Qwen 1.5 (Tongyi
 298 Qianwen License), Mistral 7B (Apache 2.0), GPT-Neo and Pythia (Apache 2.0), LLaMA 3 8B (Meta
 299 LLaMA 3 Community License), TransformerLens (MIT License). TruthfulQA and TriviaQA are used for
 300 evaluation only under their respective research licences. Released assets: HALLSCOPE (MIT License),
 301 HALLBENCH v2 (CC-BY 4.0).
- 302 13. **Assets.** *Did you document new assets?*
 303 **Yes.** HALLSCOPE (anonymised, MIT License) provides logit lens analysis, suppression ratio computation,
 304 Type 2a/2b classification, activation patching, logit blending, and HallBench v2 integration. Full API
 305 documentation is available in the GitHub repository. HALLBENCH v2 (HuggingFace: [anonymised],
 306 CC-BY 4.0) contains 50 curated examples with tier labels, first-token answers, suppression metadata, and
 307 source attribution. No personal data is included in either release.
- 308 14. **Crowdsourcing and Research with Human Subjects.** *Did you include full details of human subject*
 309 *research?*
 310 **N/A.** No crowdsourcing or human subjects research was conducted. All experiments are fully computational
 311 analyses of existing open-source language models on existing benchmark datasets.
- 312 15. **IRB Approvals.** *Did you obtain IRB approval?*
 313 **N/A.** No human subjects research was conducted. IRB approval is not applicable.
- 314 16. **Declaration of LLM Usage.** *Does the paper describe LLM usage as an important component of core*
 315 *methods?*
 316 **N/A.** Large language models are the *subject of study* in this paper, not components of the methodology. The
 317 analysis pipeline uses TransformerLens and HuggingFace Transformers for mechanistic analysis; no LLM is
 318 used to generate scientific content, write experiment code, or make analytical decisions in this work.